

Western Mathematics Exams

2021
YEAR 11
EXAMINATION

Mathematics Standard

SOLUTIONS

Multiple Choice Worked Solutions		
No	Working	Answer
1	$P(Y \text{ or } R) = \frac{13 + 16}{50} = \frac{29}{50}$ $P(\overline{Y \text{ or } R}) = 1 - \frac{29}{50} = \frac{21}{50}$	C
2	The graph is a straight line through the origin which indicates a direct variation relationship, and the equation is of the form $y = mx$ which indicates a direct variation relationship. So both are examples of direct variation	A
3	For a measurement of 6.6 kg, smallest unit = 0.1 kg Absolute error = $0.1 \div 2 = 0.05$ kg Limits of accuracy = measurement $\pm$ Absolute error So between $6.6 - 0.05$ and $6.6 + 0.05$ i.e. between 6.55 and 6.65	B
4	As all pass through the origin, they are all possible simple interest graphs Use $P = \\$800$ and $r = 0.05$ and choose a value of N between 0 – 10. If $N = 10$ $I = PRN$ $= 800 \times 0.05 \times 10 = 400$ The graph through (10, 400) is line A.	A
5	- the suburb which they last visited, is categorical, nominal data. - their body temperature is numerical, continuous data. - their age group; in classes, is categorical, ordinal data. So, the one not included is Numerical, discrete data.	D
6	$m = \frac{w}{q - p}$ $m(q - p) = w$ $mq - mp = w$ $mq = w + mp$ $q = \frac{w + mp}{m}$	D
7	Determine annual cost for each Electricity = $535.60 \times 4 = 2142.4$ Water = $213.40 \times 4 = 853.6$ Phone = $90.60 \times 12 = 1087.2$ Gas = $49.65 \times 26 = 1290.9$ Total Cost = $2142.4 + 853.6 + 1087.2 + 1290.9$ = \$5374.10 Monthly amount = $\frac{\\$5374.10}{12} = \\447.84	D

8	3:19 pm is 15:19 hours. $15:19 + 0:30 = 15:49$ This is after the second bus but before the third Time till the third bus (16:05) = 11 min + 5 min = 16 min	C
9	Weekly pay after tax = $\frac{124956}{52} \times 0.75 = \1802.25 Net pay = $1802.25 - 45.60 - 60.50 = \$1\,696.15$	A
10	Percentage less than 30 = $12.85 + 12.46 + 14.45 = 39.76\%$ Expected number, less than 30, from 80 = $0.3976 \times 80 = 31.808 = 32$ people	B

**2021 Year 11 Final Examination
Mathematics Standard**

Name _____ Teacher _____

Section I – Multiple Choice Answer Sheet

Allow about 15 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ ^{correct} C ☐ D ☐

- | | | | | |
|-----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1. | A ☐ | B ☐ | C ☒ | D ☐ |
| 2. | A ☒ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☒ | C ☐ | D ☐ |
| 4. | A ☒ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☒ |
| 6. | A ☐ | B ☐ | C ☐ | D ☒ |
| 7. | A ☐ | B ☐ | C ☐ | D ☒ |
| 8. | A ☐ | B ☐ | C ☒ | D ☐ |
| 9. | A ☒ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☒ | C ☐ | D ☐ |

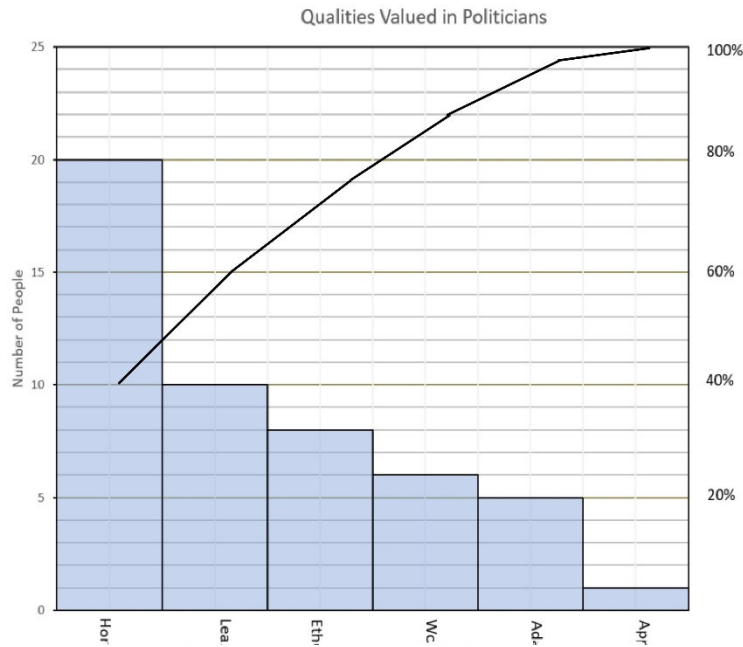
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Worked Solutions - Section II

	Solution	Marks	Allocation of marks																																				
Question 11																																							
	Time difference between NSW and SA remains half an hour So time in SA = 12:15 – 0:30 = 11:45 am Time difference between NSW and WA becomes three hours So time in WA = 12:15 – 3:00 = 9:15 am	2	2 marks for two correct answers 1 mark if only one correct																																				
Question 12																																							
(a)	$\frac{4p}{5} + 9 = 11$ $\frac{4p}{5} = 2$ $4p = 10$ $p = 2.5$	1	1 mark for correct answer																																				
(b)	$u = \frac{t^2 + mt}{r}$ $8 = \frac{2^2 + 2m}{5}$ $8 = \frac{4 + 2m}{5}$ $40 = 4 + 2m$ $36 = 2m$ $m = \frac{36}{2}$ $m = 18$	2	2 marks for correct answer 1 mark if working shown with a minor error in algebra or logic																																				
Question 13																																							
(a)	Using calculator, is most likely method or using table below <table border="1" style="margin: 10px auto; border-collapse: collapse; text-align: center;"> <thead> <tr> <th>Number of poles (<i>x</i>)</th><th>Frequency (<i>f</i>)</th><th><i>fx</i></th><th>Cumulative Frequency</th></tr> </thead> <tbody> <tr><td>12</td><td>2</td><td>24</td><td>2</td></tr> <tr><td>13</td><td>3</td><td>39</td><td>5</td></tr> <tr><td>14</td><td>2</td><td>28</td><td>7</td></tr> <tr><td>15</td><td>2</td><td>30</td><td>9</td></tr> <tr><td>16</td><td>3</td><td>48</td><td>12</td></tr> <tr><td>17</td><td>5</td><td>85</td><td>17</td></tr> <tr><td>18</td><td>3</td><td>54</td><td>20</td></tr> <tr><td colspan="2">20</td><td>308</td><td></td></tr> </tbody> </table> $\text{Mean} = \frac{308}{20} = 15.4$ Mean can also be found using the SD on calculator, without <i>fx</i> .	Number of poles (<i>x</i>)	Frequency (<i>f</i>)	<i>fx</i>	Cumulative Frequency	12	2	24	2	13	3	39	5	14	2	28	7	15	2	30	9	16	3	48	12	17	5	85	17	18	3	54	20	20		308		1	1 mark for correct answer
Number of poles (<i>x</i>)	Frequency (<i>f</i>)	<i>fx</i>	Cumulative Frequency																																				
12	2	24	2																																				
13	3	39	5																																				
14	2	28	7																																				
15	2	30	9																																				
16	3	48	12																																				
17	5	85	17																																				
18	3	54	20																																				
20		308																																					
(b)	Median = 10 th and 11 th scores and both are 16 Median = 16	1	1 mark for correct answer																																				

Question 14			
	GST Inclusive price = \$495 Grants price before GST = $495 \div 1.1 = \$450$ Suzie's price with discount = $495 \times 0.9 = \$445.50$ Suzie pays \$4.50 less for the phone	2	2 marks for correct answer 1 mark if only one of the two amounts is calculated correctly
Question 15			
(a)	Kilos burned by Tanika = $42.4 \times 48 \times 2$ = 4070 Kj	1	1 mark for correct answer
(b)	Kilos burned by Amira = $52.4 \times 65 \times \frac{20}{60}$ = 1135 Kj	1	1 mark for correct answer
Question 16			
	Depreciation in 2 years = $\$14000 - 10240 = \3760 Annual Depreciation (D) = $3760 \div 2 = 1880$ $S = V_0 - Dn$ = $14000 - 1880 \times 5$ = \$4 600 OR = $10240 - 1880 \times 3$ = \$4 600	2	2 marks for correct answers 1 mark if value of D is calculated correctly
Question 17			
(a)	Total interviews = 25 $P(\text{European}) = \frac{5}{25} = \frac{1}{5}$	1	1 mark for correct answer
(b)	$P(\text{Indigenous or African}) = \frac{5+2}{25} = \frac{7}{25}$	1	1 mark for correct answer
(c)	$P(\text{Asian}) = \frac{4}{25}$ $P(\text{not Asian}) = 1 - \frac{4}{25} = \frac{21}{25}$	1	1 mark for correct answer

Question 18

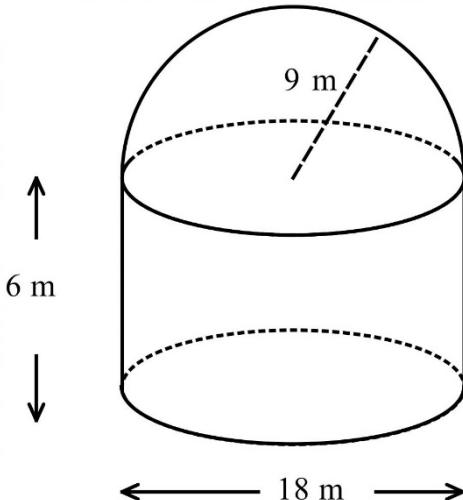
Quality	Number	Percentage	Cumulative
Honesty	20	40	40
Leadership	10	20	60
Ethics	8	16	76
Work	6	12	88
Adaptability	5	10	98
Appearance	1	2	100

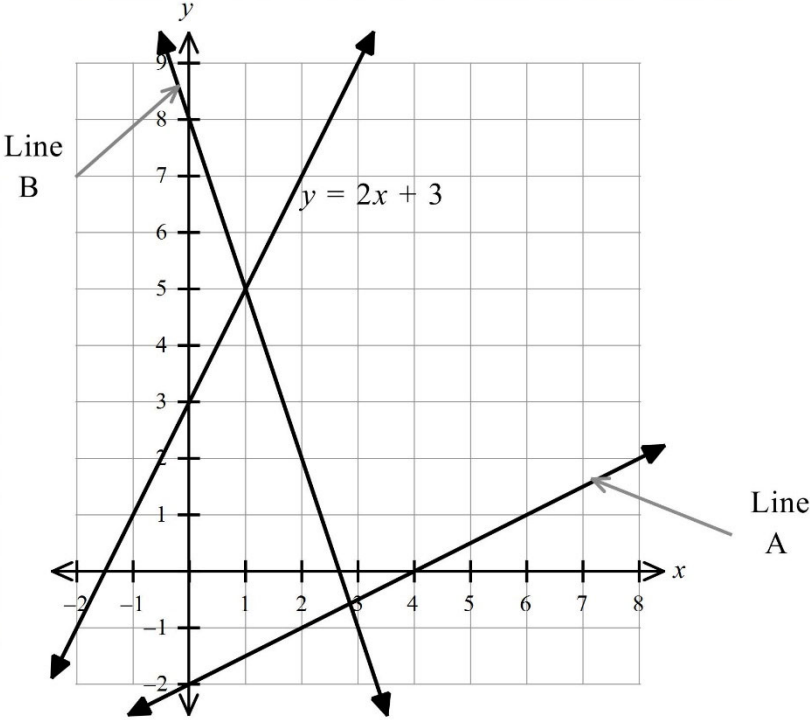


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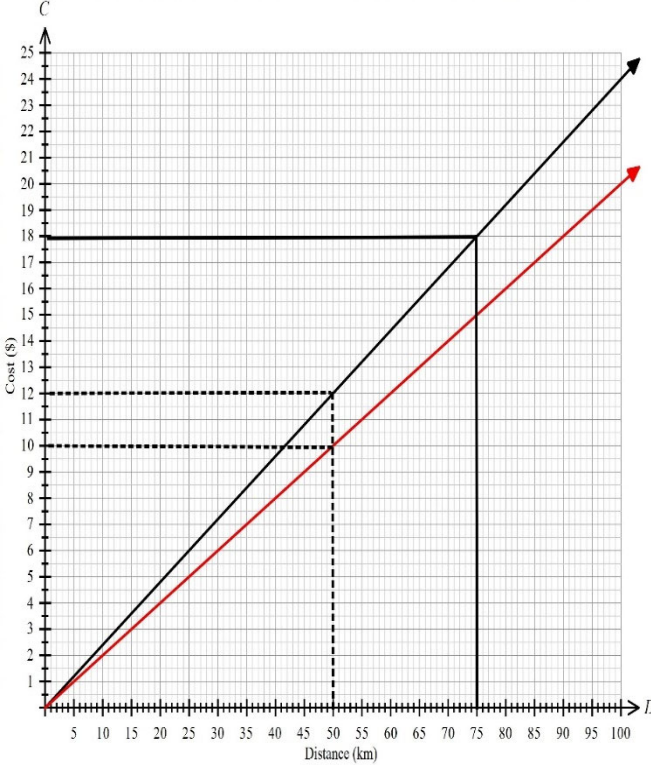
2 marks for correct graph

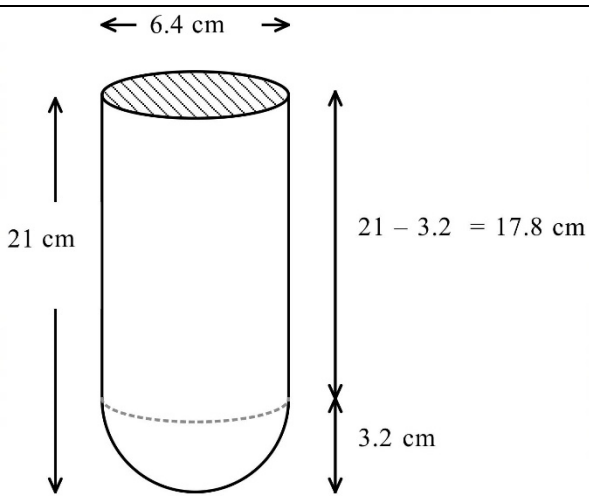
1 mark for correct percentages found or equivalent merit

Question 19			
(a)	Area Base = πr^2 = $\pi \times 9^2$ = 254.46900494 = 254 m ² (nearest m ²)	1	1 mark for correct answer
(b)	 Curved face cylinder = $2\pi r h$ = $2 \times \pi \times 9 \times 6$ = 339.24 = 339 m² (nearest m²) Curved face sphere = $4\pi r^2$ Curved face hemisphere = $2\pi r^2$ = $2 \times \pi \times 9^2$ = 508.94 = 509 m² (nearest m²) Total wall area = 339.24 + 508.94 = 848.18 = 848 m² (nearest m²)	2	2 marks for correct answer

Question 21			
(a)	Using two points eg (0, -2) and (4, 0) Gradient = $\frac{\text{rise}}{\text{run}} = \frac{2}{4} = \frac{1}{2}$	1	1 mark for correct answer
(b)	Gradient = $\frac{\text{rise}}{\text{run}} = -\frac{3}{1} = -\frac{6}{2} = -3$ y – intercept = 8 Equation : $y = -3x + 8$	1	1 mark for correct answer
(c)	 Point of intersection (1,5)	2	2 marks for correct graph and point 1 mark if only one of the two is completed correctly
Question 22			
	$\text{Area} = \frac{40}{2}(90 + 100) + \frac{40}{2}(100 + 65)$ $= 3800 + 3300$ $= 7100 \text{ m}^2$	2	2 marks for correct answer 1 mark for working that shows some relevant area calculations

Question 27			
(a)	$\text{Susan } BAC_{FEMALE} = \frac{10 \times 3 - 7.5 \times 2}{5.5 \times 60} = \frac{15}{330} = 0.04545455$ $= 0.045$ $\text{Adam } BAC_{MALE} = \frac{10 \times 3 - 7.5 \times 2}{6.8 \times 90} = \frac{15}{612} = 0.0245098$ $= 0.025$	2	2 marks for correct answers 1 mark for one correct answer, or similar simple error in both answers Don't deduct mark for rounding
(b)	$\text{Time}_{Susan} = \frac{0.04545455}{0.015} = 3.03030333$ $\text{Time}_{Adam} = \frac{0.0245098}{0.015} = 1.63398667$ $\text{Difference} = 3.03030333 - 1.63398667 = 1.39631666$ $= 1 \text{ hour and 24 minutes}$ [accept 1 hour and 20 min- answer using rounded values]	1	1 mark for correct answer
Question 28			
(a)	$\text{Taxable income} = 98500 - (460 + 780 + 3500)$ $= \$93760$ $\text{Tax on } \$93760 = (\$93760 - 45000) \times 0.325 + 5092$ $= \$20939$	2	2 marks for correct answer 1 mark for correct taxable income or equivalent merit
(b)	$\text{Total Tax Due} = \$20939 + 1875.20 = \$22\,814.20$ $\text{Total PAYG} = 880.50 \times 26 = \$22\,893.00$ $\text{Refund} = 22893.00 - 22814.20 = \78.80 She will get a \$78.80 refund.	1	1 mark for correct answer
Question 29			
(a)	$P(A \text{ and } N) = 0.6 \times 0.5 = 0.3$	1	1 mark for correct answer
(b)	$P(S \text{ or } O) = P(SM) + P(SN) + P(SO) + P(AO)$ $= 0.4 \times 0.3 + 0.4 \times 0.5 + 0.4 \times 0.2 + 0.6 \times 0.2$ $= 0.52$ OR: $P(S \text{ or } O) = P(S) + P(AO) = 0.4 + 0.12 = 0.52$	2	2 marks for correct answer 1 mark for working that shows some relevant probabilities
(c)	$P(\text{not } A \text{ or } M) = P(SN) + P(SO)$ $= 0.4 \times 0.5 + 0.4 \times 0.2$ $= 0.28$	1	1 mark for correct answer
Question 30			
	Find length of diameter of larger semi – circle d $d^2 = 8^2 + 15^2$ $= 289$ $d = \sqrt{289} = 17$ length of larger semicircle $= 2 \times \pi \times 8.5 \div 2 = 26.70353756$ length of smaller semicircle $= 2 \times \pi \times 5 \div 2$ $= 15.70796327$ $\text{Perimeter} = 15 + 18 + 15.7079 + 26.7035 = 75.4114$ $= 75.4 \text{ m (nearest 10th metre)}$	2	2 marks for correct answer 1 mark for calculating diameter correctly and using incorrectly or equivalent merit

Question 31			
(a)	Childs dosage = $\frac{6 \times 7.5}{6 + 12} = 2.5 \text{ ml}$	1	1 mark for correct answer
(b)	Adult dosage = $2 \times 28 = 56 \text{ mg}$ Childs dosage = $\frac{8 \times 56}{8 + 12} = 22.4 \text{ mg} = 22 \text{ mg (nearest mg)}$ Half a 28 mg tablet = 14 mg $14 \text{ mg} + 8 \text{ mg} = 22 \text{ mg}$ She would receive half of a 28 mg tablet and a full 8 mg tablet	2	2 marks for correct answer 1 mark for calculating dosage in mg correctly or equivalent merit
Question 32			
(a)	Stamp duty = $1350 + \$5 \times 558$ $= \$1350 + 2790 = \$4\ 140$	1	1 mark for correct answer
(b)	Comp Ins = $1250 \times 0.8 = \$1000.00$	1	1 mark for correct answer
(c)	Total on road costs = $\$675.50 + 4140 + 342.50 + 1000$ $= \$6158$ Total cost of car = $\$55730 + 6158 = \$61\ 888.00$	1	1 mark for correct answer
Question 33			
(a)	 Cost = \$18: See unbroken line above.	1	1 mark for correct answer
(b)	Equation is $C = 0.2D$ R line on the graph above	1	1 mark for correct graph
(c)	Using broken lines, Lucy's costs \$12 and Ayla's costs \$10, so it costs \$2 more	1	1 mark for correct answer

Question 34			
(a)	Weekly rent = $\$900 \div 2 \div 2 = \225.00 Weekly energy = $485 \times 4 \div 52 \div 2 = \18.65 Total weekly account = $225 + 18.65 = \$243.65$	1	1 mark for correct answer
(b)	Annual Personal = $1859 \times 12 = \$22308$ Weekly Personal = $22308 \div 52 = \$429$	1	1 mark for correct answer
(c)	Total Ent and Car = $68 + 108 = 176$ Total with rent and personal = $176 + 243.65 + 429 = \$848.65$ Weekly saving amount = $880 - 848.65 = \$31.35$	1	1 mark for correct answer
Question 35			
(a)	 Volume of hemisphere = $\frac{4}{3} \times \pi \times 3.2^3 \div 2$ $= 68.629 \text{ cm}^3$ Volume of cylinder = $\pi \times 3.2^2 \times 17.8$ $= 572.624 \text{ cm}^3$ Total volume = $572.624 + 68.629$ $= 641.253$ $= 641 \text{ cm}^3$ nearest cm^3	2	2 marks for correct answer 1 mark for an incorrect volume calculated using incorrect radius or height or equivalent merit
(b)	Capacity is 641 mL Each press uses $0.02\text{L} \times 1000 = 20 \text{ mL}$ Number of uses = $641 \div 20 = 32.05$ So it would need 33 presses to empty the container.	1	1 mark for correct answer calculated from answer to part (a)

Question 36			
(a)	Jennifers difference to Samantha $= +10 + 3 = +13$ hours Jennifer is 13 hours ahead	1	1 mark for correct number of hours
(b)	Jennifers difference to Marissa $= +10 \mp 2 = +8$ hours Marissa is 8 hours behind 4 pm, so 8 am	1	1 mark for correct answer
(c)	Arrives 10 hours after 12 noon on 15th August $= 10$ pm Brasillia time Jennifer is 13 hours ahead, so 13 ahead of 10 pm $= 11$ am on 16th August	2	2 marks for correct answer 1 mark for correct arrival time or equivalent merit